UX RESEARCH · AIS PAY & TOP-UP

UX Research for Pay & Top-up

Research plan, proto-personas and hypotheses that shaped the 35 screens — to be validated with real users.

Status · Plan & hypotheses · 8 interviews · 6 tests · 4 weeks

01 Context

PROBLEM STATEMENT

AIS customers who pay bills and top up for themselves and their family have to jump between apps, bank transfers and counters. They can't easily see what they're charged for, whether a payment worked, or whose number needs money.

RESEARCH GOALS

- Understand how people pay bills and top up today — channels, triggers, frequency
- Find where trust breaks: wrong numbers, unclear charges, failed payments
- Validate that one app for self + family numbers is wanted
- Test the prototype's top-up and bill-pay flows for speed and clarity

RESEARCH QUESTIONS

- RQ1 · What triggers a top-up or bill payment, and who decides the amount?
- RQ2 · How do people confirm a payment succeeded today?
- RQ3 · Which bill lines cause confusion or calls to support?
- RQ4 · How many numbers does a household manage, and how?
- RQ5 · What would make people trust auto-pay?

02 Methods & timeline

Desk research & channel audit — Week 1

Review app-store reviews, support FAQs and existing payment channels.

Output · Pain-point list · channel comparison

In-depth interviews — Week 1–2

8 × 45-min remote sessions across 3 segments.

Output · Quotes · journey evidence · personas

Quick survey — Week 2

≈150 responses on channels, frequency, family numbers and auto-pay trust.

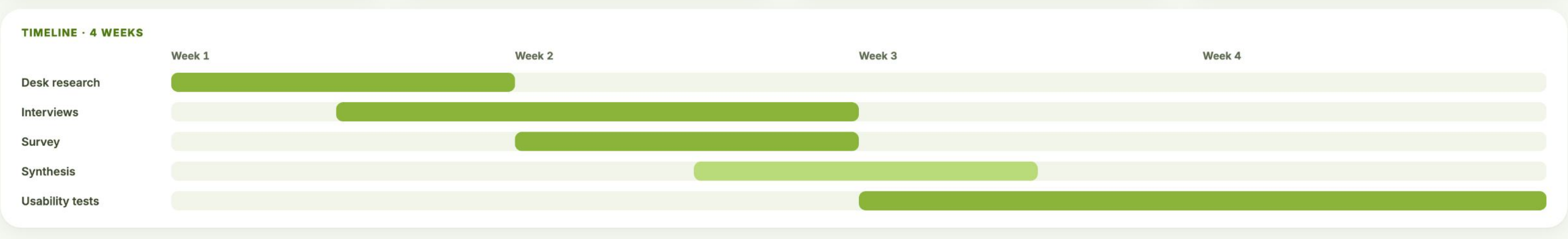
Output · Sizing of key behaviours

Usability testing — Week 3–4

6 sessions on the clickable prototype, 5 tasks each.

Output · Task success · time · SUS score

TIMELINE · 4 WEEKS



03 Who we'll talk to

Recruitment and proto-personas

RECRUITMENT · 8 PARTICIPANTS

Postpaid self-payers
3 interviews · 2 tests
Pay their own monthly bill; 25–45; mostly PromptPay or card.

Family top-up managers
3 interviews · 2 tests
Top up 2+ prepaid numbers for parents, kids or staff.

Light / older users
2 interviews · 2 tests
45+; use counters or ask family; lower app confidence.

Screener: active AIS number, paid or topped up in the last 30 days; mix of Bangkok and regional users.

S Somchai, 34
Office worker · Postpaid
GOAL
Pay on time with no effort; know exactly what he's charged for
FRUSTRATION
Unexplained extra charges; unsure payments went through
"If it says paid, I want proof I can show."

M Malee, 52
Shop owner · tops up 3 numbers
GOAL
Keep parents' and staff phones working without visiting a counter
FRUSTRATION
Typing numbers wrong; forgetting whose balance is low
"I just want to see who needs money this week."

P Ploy, 22
Student · Prepaid, heavy data
GOAL
Buy the cheapest data pack that lasts until the next top-up
FRUSTRATION
Running out of data mid-month; confusing pack names
"Tell me how many days my data will last."

04 Channel comparison

How people pay today vs. our concept

Capability	Carrier app (today)	Banking app	E-wallet	Counter / kiosk	Pay & Top-up (ours)
Top up another person's number	Partial	Partial	Partial	Partial	Yes
Checks number is AIS before paying	Partial	—	—	Partial	Yes
Itemised bill with reasons for changes	Partial	—	—	—	Yes
Saved family numbers with balances	—	Partial	Partial	—	Yes
Auto-pay with reminder before charging	Partial	Partial	—	—	Yes
Receipts kept in one place	Partial	Partial	Partial	—	Yes
Clear recovery when a payment fails	—	Partial	—	Partial	Yes

Ratings are working assumptions from desk research — confirm each one during the channel audit (week 1).

05 Insight clusters

Hypotheses from desk research — validate in interviews

Trust & proof
- Users screenshot bank slips because the carrier status updates late
- A clear "Paid" state plus a reference number reduces support calls
- Failed payments feel scary when no next step is shown

› Design response · Receipts (21) · Bill paid (19) · Payment failed (20)

Speed for repeat tasks
- Most top-ups go to the same 2–3 numbers
- Users want one tap from Home to a saved person
- PIN or biometrics is fine if it replaces OTP every time

› Design response · Home quick top-up (08) · Saved numbers (32) · Biometrics (07)

Bill clarity
- "Why is it higher this month?" is the top bill question
- People want add-ons shown with dates, not one total
- A month-to-month comparison builds confidence

› Design response · Pay bill insight (16) · Invoice (17)

Control over automation
- Auto-pay is trusted only with a reminder and a cap
- Users want to see the next charge date and amount
- Easy to switch off is a must before switching on

› Design response · Auto-pay (22) · Notifications (30)

06 Usability test & interview guide

USABILITY TEST · 5 TASKS ON THE PROTOTYPE

	Task	Screens	Target
T1	Top up ฿100 to Mom's number	Screens 09 → 12 → 14 → 15	≥ 90% success · < 45 s
T2	Find why this month's bill is higher	Screens 08 → 16 → 17	≥ 80% find the 5 GB pack
T3	Pay the bill with PromptPay QR	Screens 16 → 18 → 19	≥ 90% success · no help
T4	Recover from a declined card	Screens 20 → 18	≥ 80% choose a working method
T5	Turn on auto-pay with a ฿1,000 cap	Screens 08 → 22	≥ 80% success · trust rating ≥ 4/5

SUCCESS METRICS

Task success	Time on task	SUS score	Errors per task
≥ 85%	Top-up < 45 s	≥ 75	≤ 1

INTERVIEW GUIDE · CORE QUESTIONS

- Q1 Walk me through the last time you topped up or paid a phone bill.
- Q2 How did you know the payment went through?
- Q3 Tell me about a time a bill amount surprised you.
- Q4 Whose numbers do you look after? How do you keep track?
- Q5 What would you need to see before turning on auto-pay?
- Q6 If this app disappeared tomorrow, what would you miss?

45 min · consent & warm-up 5 · behaviour 20 · concept 10 · wrap-up 10

UX RESEARCH · AIS PAY & TOP-UP

Personas for Pay & Top-up

Three proto-personas built from desk research. Validate and update them after the 8 interviews.

Proto-personas · to validate

SPEED
Pay or top up in under 60 seconds
→ Quick top-up, saved numbers, biometrics

CLARITY
Know exactly what you pay for
→ Itemised invoice, "more than last month" insight

TRUST
Proof that money arrived
→ Success receipts, reference numbers, Receipts page

CONTROL
Automation without surprises
→ Auto-pay cap, reminder before charging, alerts

S Somchai Jaidee

PRIMARY PERSONA
The Busy Bill-Payer

Age	Location	Job	Plan
34	Bangkok	Office worker	AIS Postpaid ฿599

Works long hours and handles everything on his phone. Pays his own postpaid bill each month and sometimes helps his mum with her prepaid number. Hates surprises on his bill and wants proof when something is paid.

> *"If it says paid, I want a receipt I can show."*

GOALS
- Pay the bill on time in under a minute
- Understand every line on the invoice
- Set up auto-pay once and forget about it

FRUSTRATIONS
- Bill went up ฿45 with no explanation
- Bank app says paid, carrier still shows unpaid
- OTP every single time feels slow

BEHAVIOURS
- Pays around the 20th, after salary
- Uses PromptPay QR from his banking app
- Screenshots slips "just in case"

CHANNELS TODAY
Mobile banking · PromptPay QR · Carrier app

Scale	Score
Tech comfort (Low–High)	5/5
Price sensitivity (Low–High)	3/5
Need for control (Low–High)	4/5

SCENARIO
Gets a bill reminder on the 20th, notices it's higher than last month, checks the invoice on the train and pays by QR before reaching the office.

KEY SCREENS FOR THIS PERSONA
08 Home · 16 Pay bill · 17 Invoice · 18 Pay with QR · 22 Auto-pay

M Malee Somboon

SECONDARY PERSONA
The Family Top-up Manager

Age	Location	Job	Numbers
52	Khon Kaen	Shop owner	3 prepaid + own

Runs a small grocery and keeps her parents' and two staff members' phones topped up. Used to buy top-up cards at the counter; now tries to do it on her phone but worries about sending money to the wrong number.

> *"I just want to see who needs money this week."*

GOALS
- Top up the right person in a few taps
- See everyone's balance in one place
- Get warned before a number runs out

FRUSTRATIONS
- Typing 10-digit numbers by hand
- Forgetting whose balance is low
- Unsure if a top-up actually arrived

BEHAVIOURS
- Tops up ฿50–100 several times a month
- Calls family to ask if credit arrived
- Prefers big buttons and clear labels

CHANNELS TODAY
Counter / 7-Eleven · Mobile banking · LINE with family

Scale	Score
Tech comfort (Low–High)	2/5
Price sensitivity (Low–High)	4/5
Need for reassurance (Low–High)	5/5

SCENARIO
Her mother calls saying the phone has no credit. Malee opens the app, taps Mom in Quick top-up, picks ฿100, confirms with her fingerprint and sees "Mom's balance ฿135.50".

KEY SCREENS FOR THIS PERSONA
08 Quick top-up · 09 Top up · 10 Contacts · 15 Top-up success · 32 Saved numbers

P Ploy Chaiyaporn

SECONDARY PERSONA
The Data-Hungry Student

Age	Location	Job	Plan
22	Chiang Mai	University student	AIS Prepaid

Streams, games and studies on mobile data. Budget is tight, so she buys short data packs when she runs low. Often confused by pack names and how long the data will really last.

> *"Tell me how many days my data will last."*

GOALS
- Never run out of data mid-month
- Pick the best-value pack fast
- Track what she spends each month

FRUSTRATIONS
- Pack names don't say what she gets
- Data runs out without warning
- Hard to compare price per GB

BEHAVIOURS
- Buys a pack 2–3 times a month
- Checks usage almost daily
- Pays with an e-wallet or PromptPay

CHANNELS TODAY
E-wallet · Carrier app · USSD codes

Scale	Score
Tech comfort (Low–High)	5/5
Price sensitivity (Low–High)	5/5
Data usage (Low–High)	5/5

SCENARIO
Gets a "85% data used" alert during exams, opens Usage, sees 8.8 GB left with 22 days to go, compares packs and buys 10 GB for ฿199.

KEY SCREENS FOR THIS PERSONA
27 Usage · 23 Data packs · 24 Package detail · 26 Package active · 28 History

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UX RESEARCH · AIS PAY & TOP-UP

Customer journey map

Hypothesis map — validate quotes, emotions and pain points with user interviews before final design.

PROTO-PERSONA
S Somchai, 34
Office worker in Bangkok. Pays his own AIS postpaid bill and tops up prepaid numbers for his mother and sister.

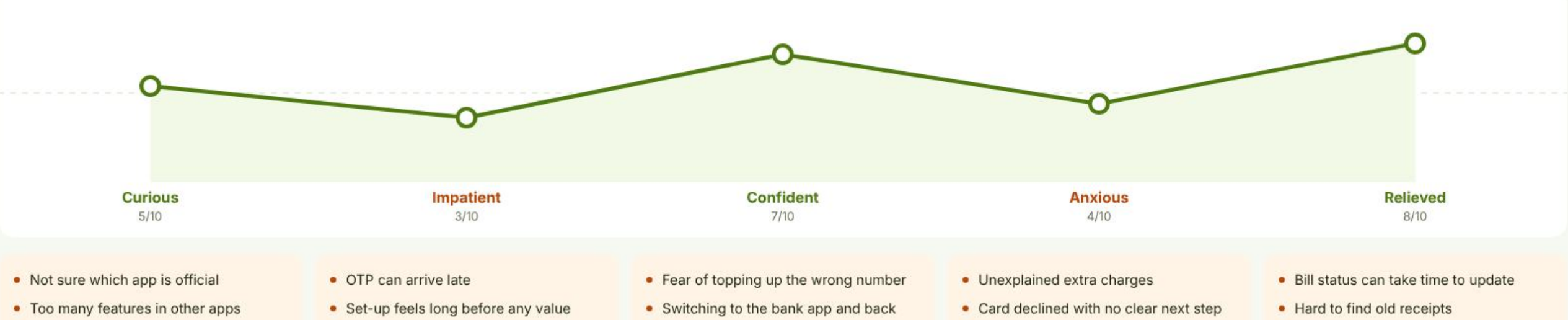
Goals
- Pay on time with as few taps as possible
- Keep his family's numbers working
- Know exactly what he's charged for

Frustrations
- Hard-to-find menus in big apps
- Not sure a payment went through
- Typing phone numbers by hand

Scenario
It's the 20th. Somchai gets a bill reminder, notices Mom's number is almost out of credit, and wants to sort both out during his commute.

Stage	1 Discover & install	2 Sign in & set up	3 Top up family	4 Pay monthly bill	5 After payment & support
User goal	Find a faster way to pay than going to a shop or counter	Get in quickly and know the account is secure	Refill Mom's prepaid number before it runs out	Understand the bill and pay it on time	Have proof of payment and stop worrying
Actions What the user does	Gets an SMS bill reminder Searches the app store Installs and opens the app	Enters mobile number Waits for and types the OTP Creates a 6-digit PIN Enables biometrics	Picks Mom from saved numbers or contacts Chooses ฿100 Pays with PromptPay Gets a confirmation	Opens the bill from a reminder Checks the itemised charges Pays by QR or card Confirms with PIN	Saves or shares the receipt Turns on auto-pay Checks history later Contacts support if needed
Touchpoints	SMS App store Friends & family	SMS OTP App Phone biometrics	Contacts Banking app Push notification	Push reminder Invoice PDF Banking app	Receipt History Chat / call centre
Thinking	*"Is there an app that just does bills and top-ups?"*	*"Why do I need an OTP and a PIN?"*	*"Did I type her number right?"*	*"Why is it ฿45 more than last month?"*	*"Good, it's done and I have the slip."*

Feeling
Emotional high and low points



Stage	1 Discover & install	2 Sign in & set up	3 Top up family	4 Pay monthly bill	5 After payment & support
Pain points	Not sure which app is official Too many features in other apps	OTP can arrive late Set-up feels long before any value	Fear of topping up the wrong number Switching to the bank app and back	Unexplained extra charges Card declined with no clear next step	Bill status can take time to update Hard to find old receipts
Opportunities How the design responds	Short value-first onboarding Clear "for AIS numbers" message → Screens 01–03	OTP auto-fill and visible resend timer Explain why PIN protects payments Make biometrics skippable → Screens 04–07	Live AIS-network check on the number Saved numbers with nicknames Review step before paying → Screens 09–15	Line-by-line invoice with add-on dates Clear failure screen with retry options "More than last month" insight on the bill → Screens 16–22	Status timeline on every transaction Receipts kept in History In-app chat with transaction context → Screens 28–35
Success metric	Install → sign-in rate	Time to finish set-up	Top-up success rate	On-time payment rate	Auto-pay adoption

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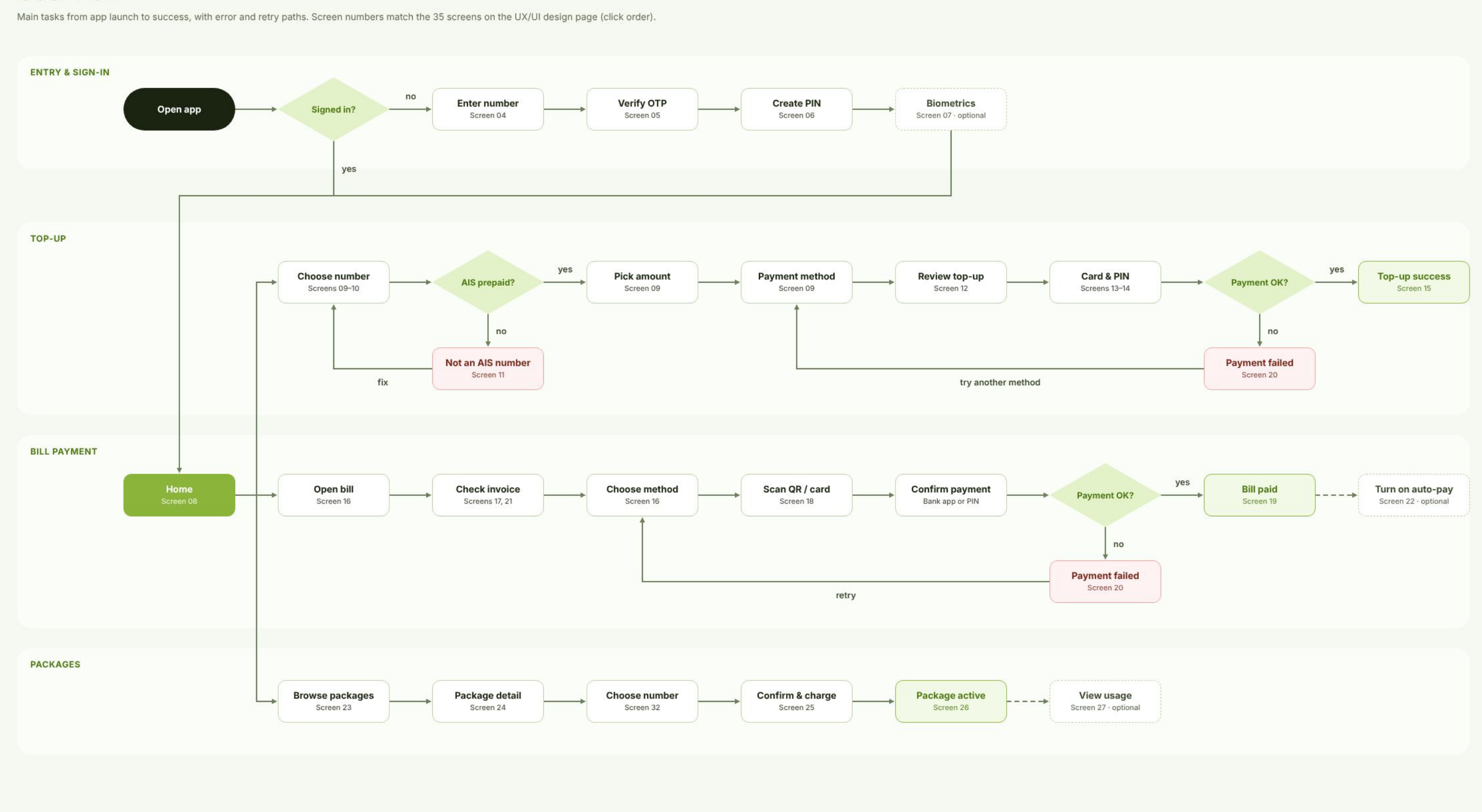
UX · AIS PAY & TOP-UP

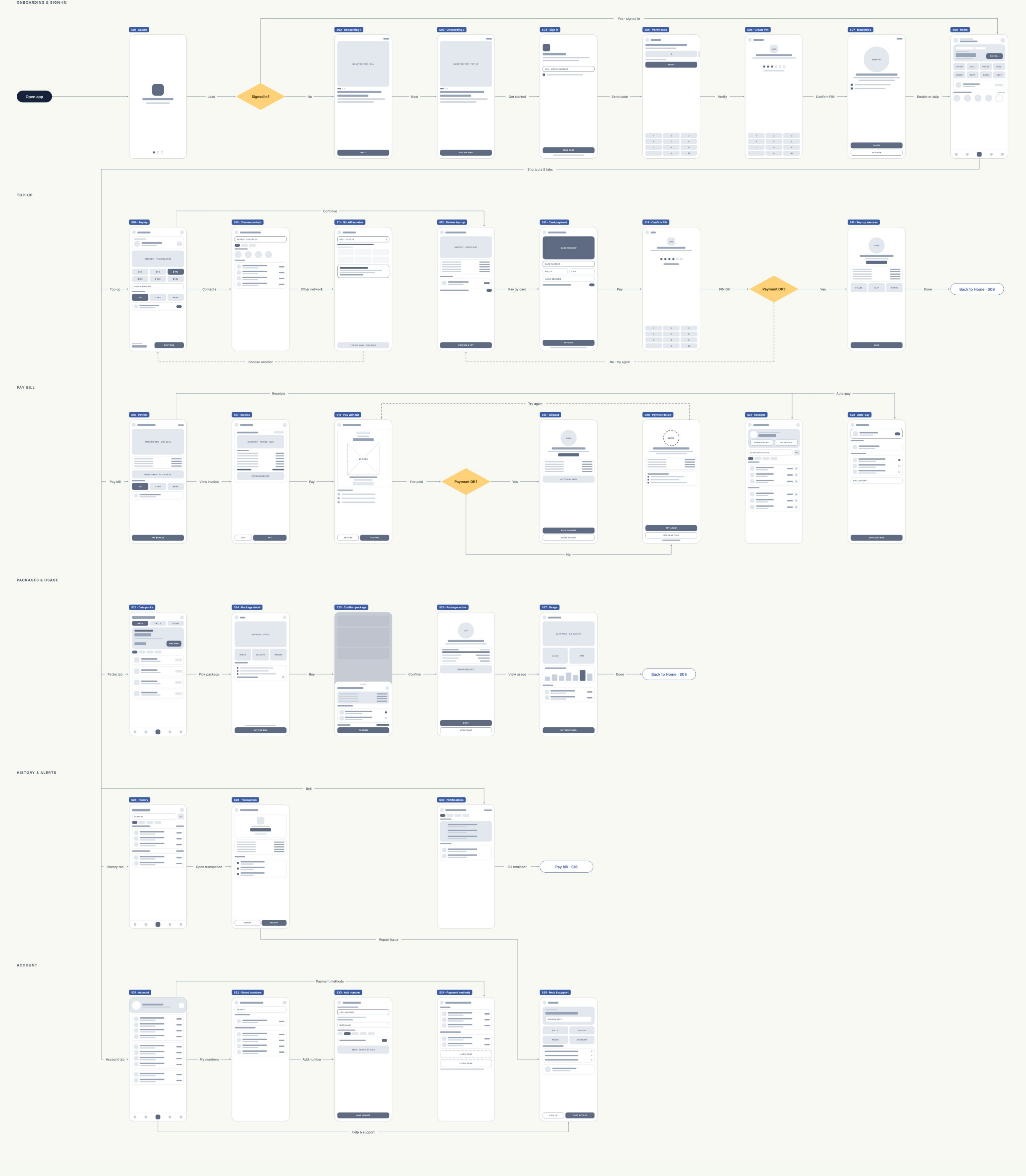
User flow

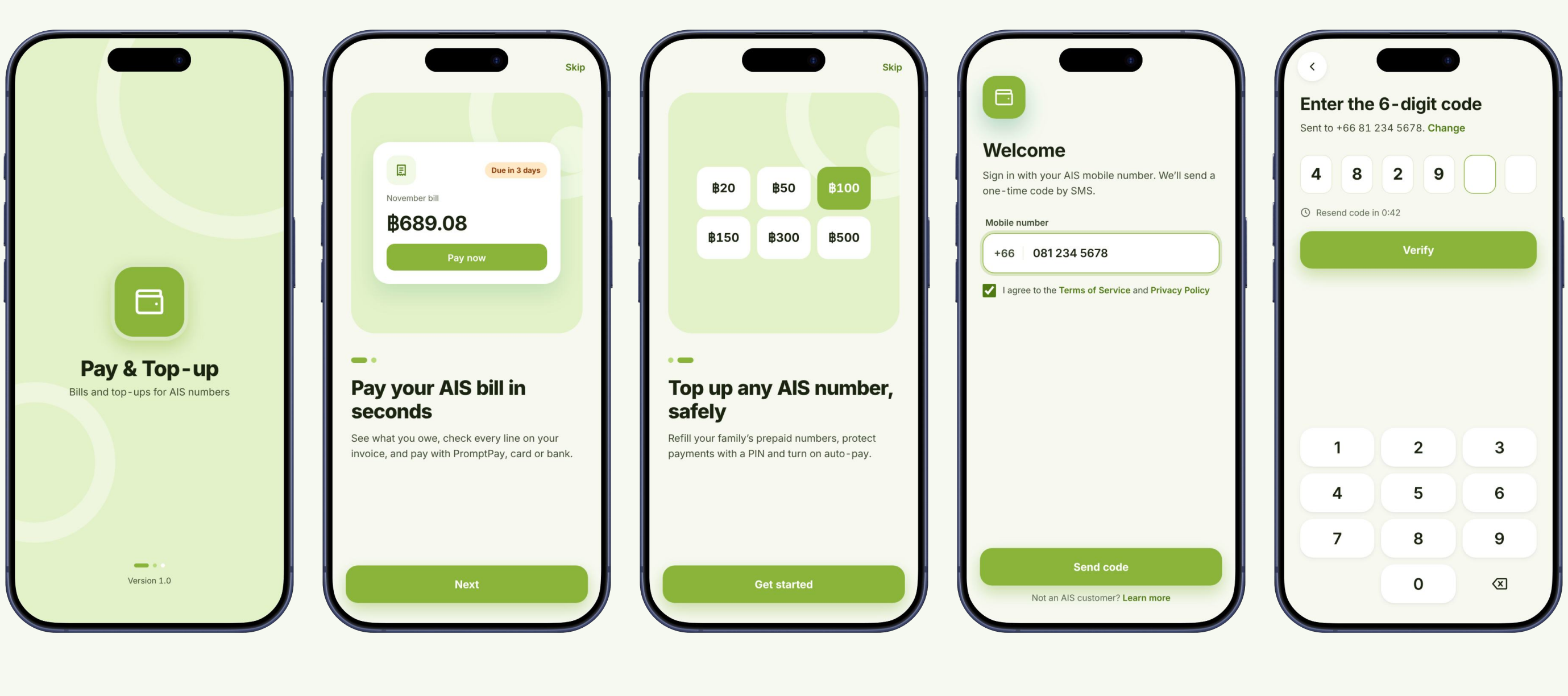
Main tasks from app launch to success, with error and retry paths. Screen numbers match the 35 screens on the UX/UI design page (click order).

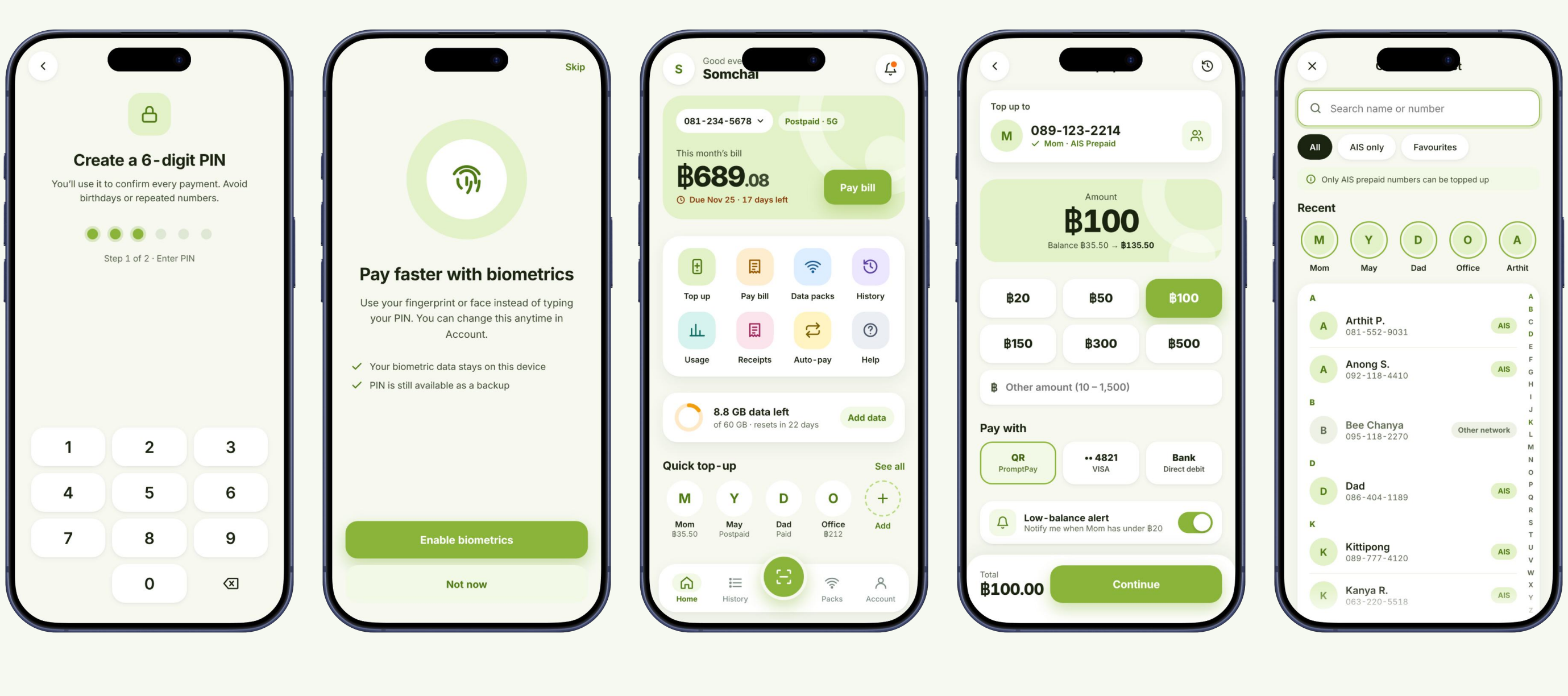

LEGEND
Start
Screen
Hub
Success
Error
Optional
?
ENTRY & SIGN-IN
Open app
Signed in?
no
yes
Enter number
Screen 04
Verify OTP
Screen 05
Create PIN
Screen 06
Biometrics
Screen 07 · optional
TOP-UP
Choose number
Screens 09–10
AIS prepaid?
yes
no
Not an AIS number
Screen 11
fix
Pick amount
Screen 09
Payment method
Screen 09
Review top-up
Screen 12
Card & PIN
Screens 13–14
Payment OK?
yes
no
Top-up success
Screen 15
Payment failed
Screen 20
try another method
BILL PAYMENT
Home
Screen 08
Open bill
Screen 16
Check invoice
Screens 17, 21
Choose method
Screen 16
Scan QR / card
Screen 18
Confirm payment
Bank app or PIN
Payment OK?
yes
no
Bill paid
Screen 19
Turn on auto-pay
Screen 22 · optional
Payment failed
Screen 20
retry
PACKAGES
Browse packages
Screen 23
Package detail
Screen 24
Choose number
Screen 32
Confirm & charge
Screen 25
Package active
Screen 26
View usage
Screen 27 · optional

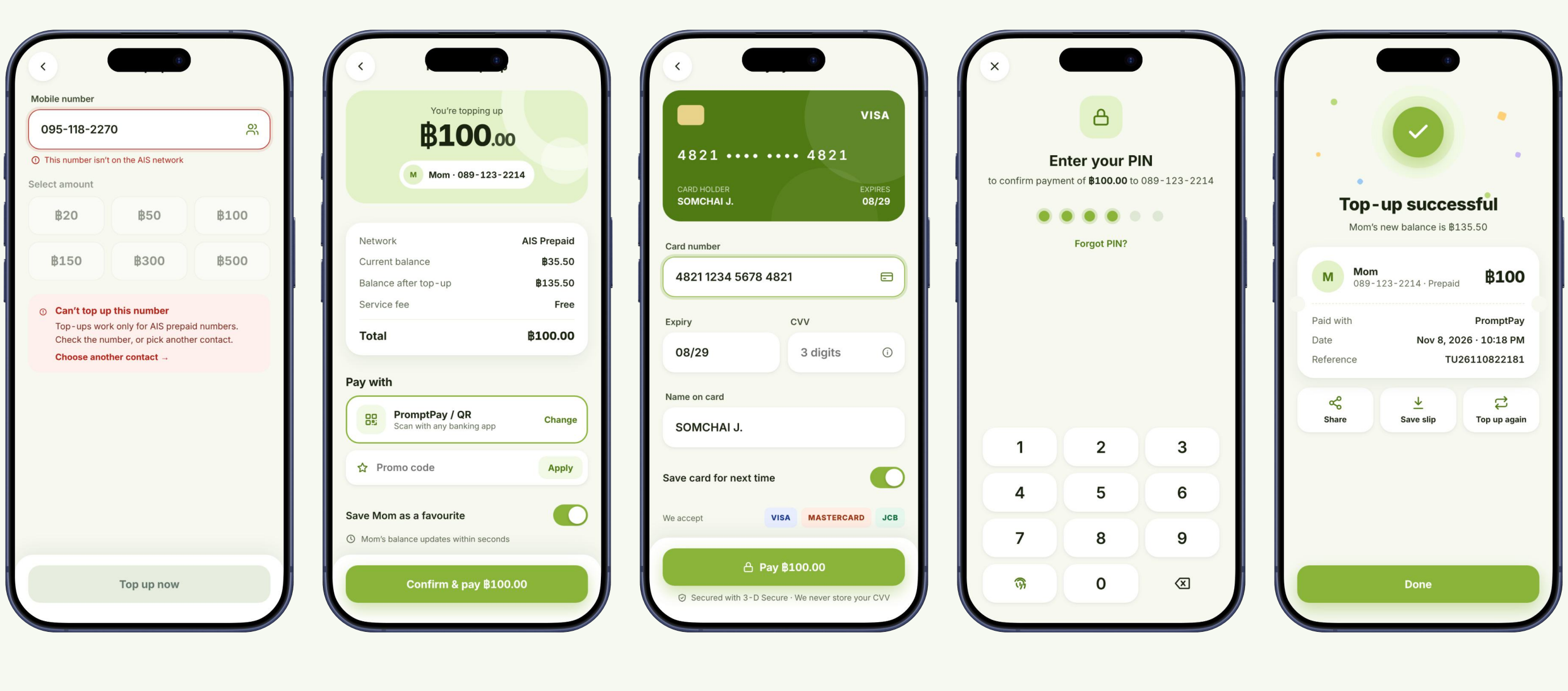
Wireframe User Flow — Pay & Top-up

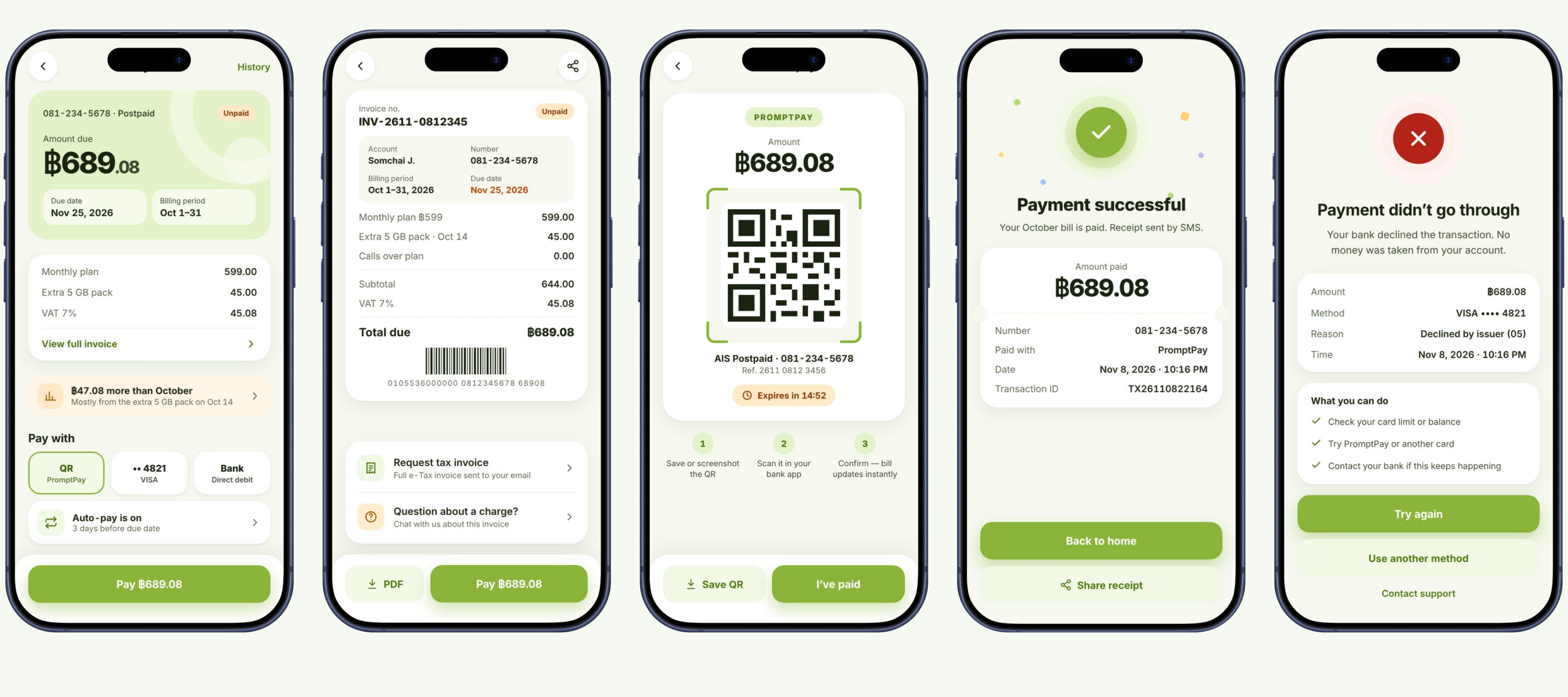
AIS bill-pay app · 35 low-fidelity wireframes connected by tap paths: onboarding & sign-in, top-up, bill payment, packages, history and account. Badges show the screen number used on the UX/UI design page (click order).

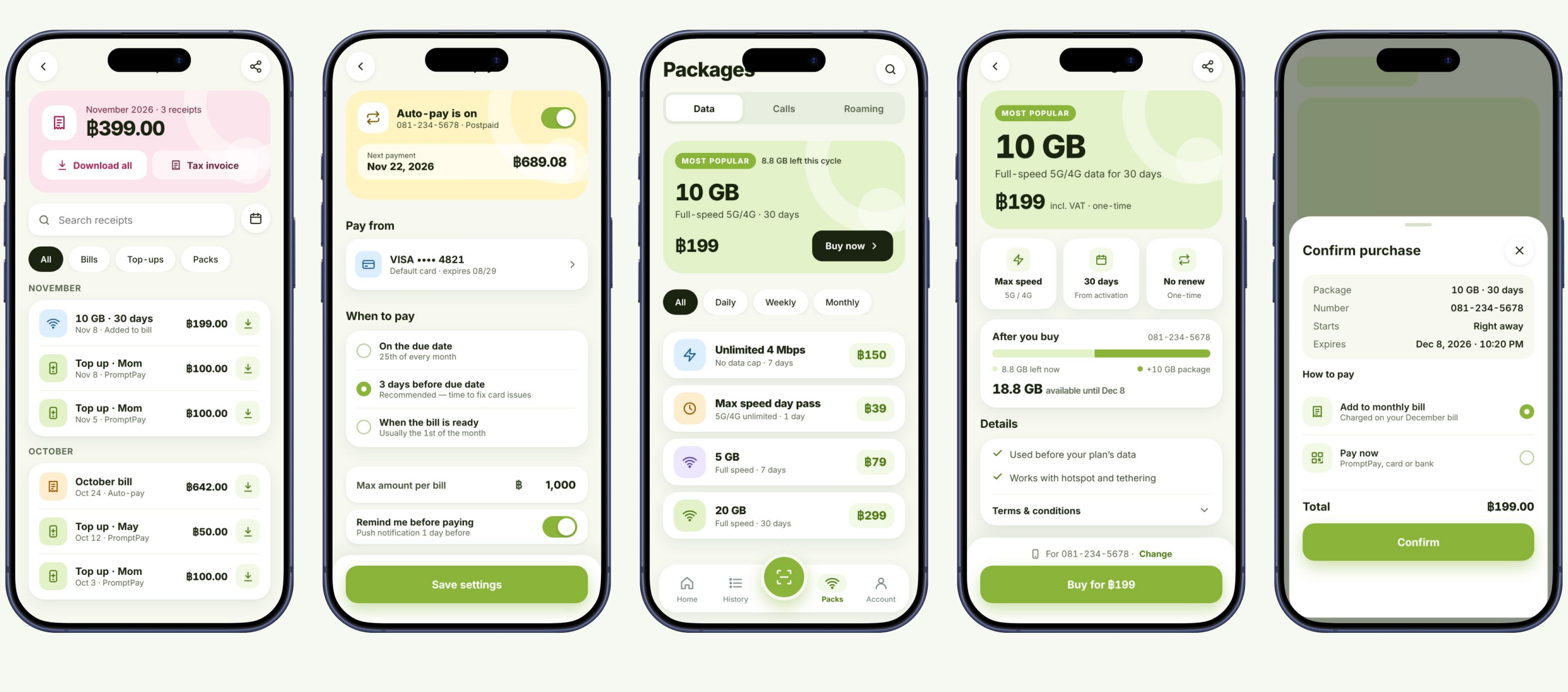

ONBOARDING & SIGN-IN
Open app
Signed in?
TOP-UP
ACCOUNT

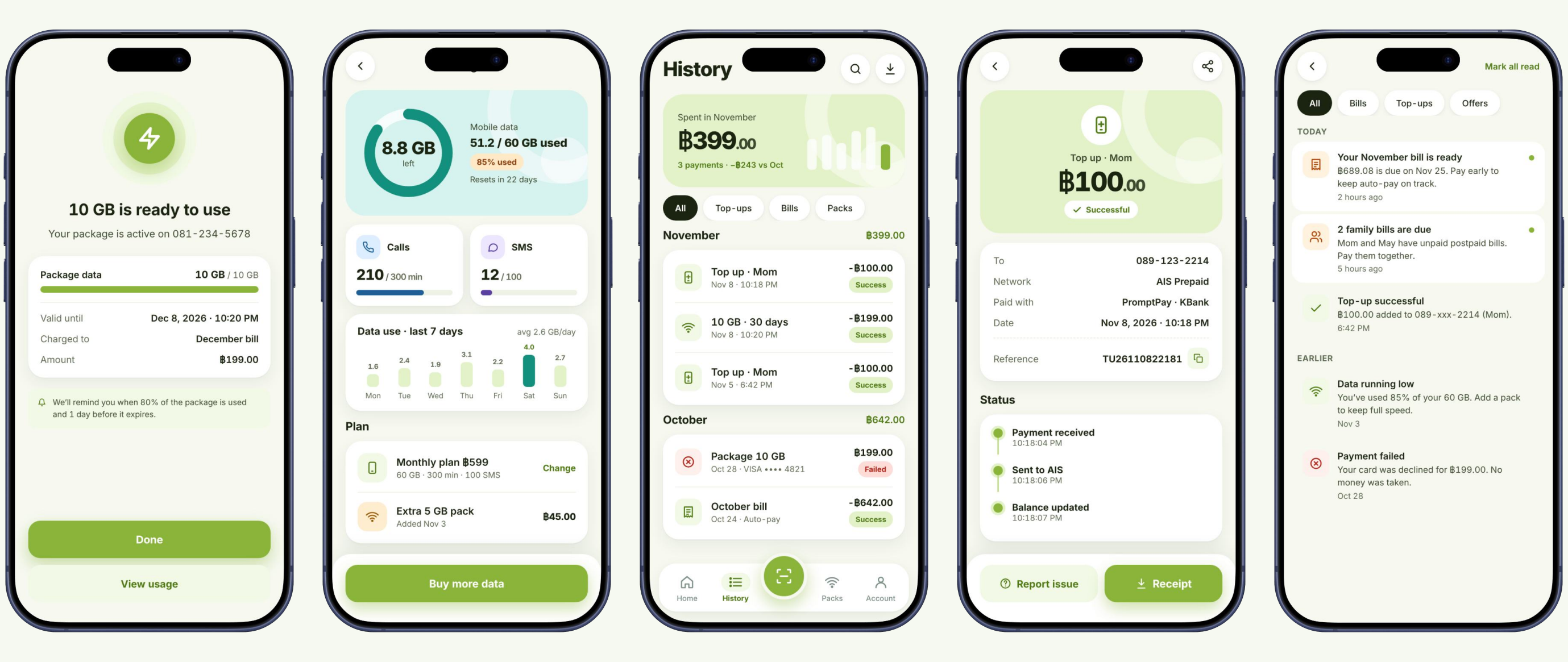

Pay & Top-up
Bills and top-ups for AIS numbers
Version 1.0
Skip
Due in 3 days
November bill
฿689.08
Pay now
Pay your AIS bill in seconds
See what you owe, check every line on your invoice, and pay with PromptPay, card or bank.
Next
฿20 ฿50 ฿100 ฿150 ฿300 ฿500
Top up any AIS number, safely
Refill your family's prepaid numbers, protect payments with a PIN and turn on auto-pay.
Get started
Welcome
Sign in with your AIS mobile number. We'll send a one-time code by SMS.
Mobile number
+66 081 234 5678
I agree to the Terms of Service and Privacy Policy
Send code
Not an AIS customer? Learn more
Enter the 6-digit code
Sent to +66 81 234 5678. Change
Resend code in 0:42
Verify

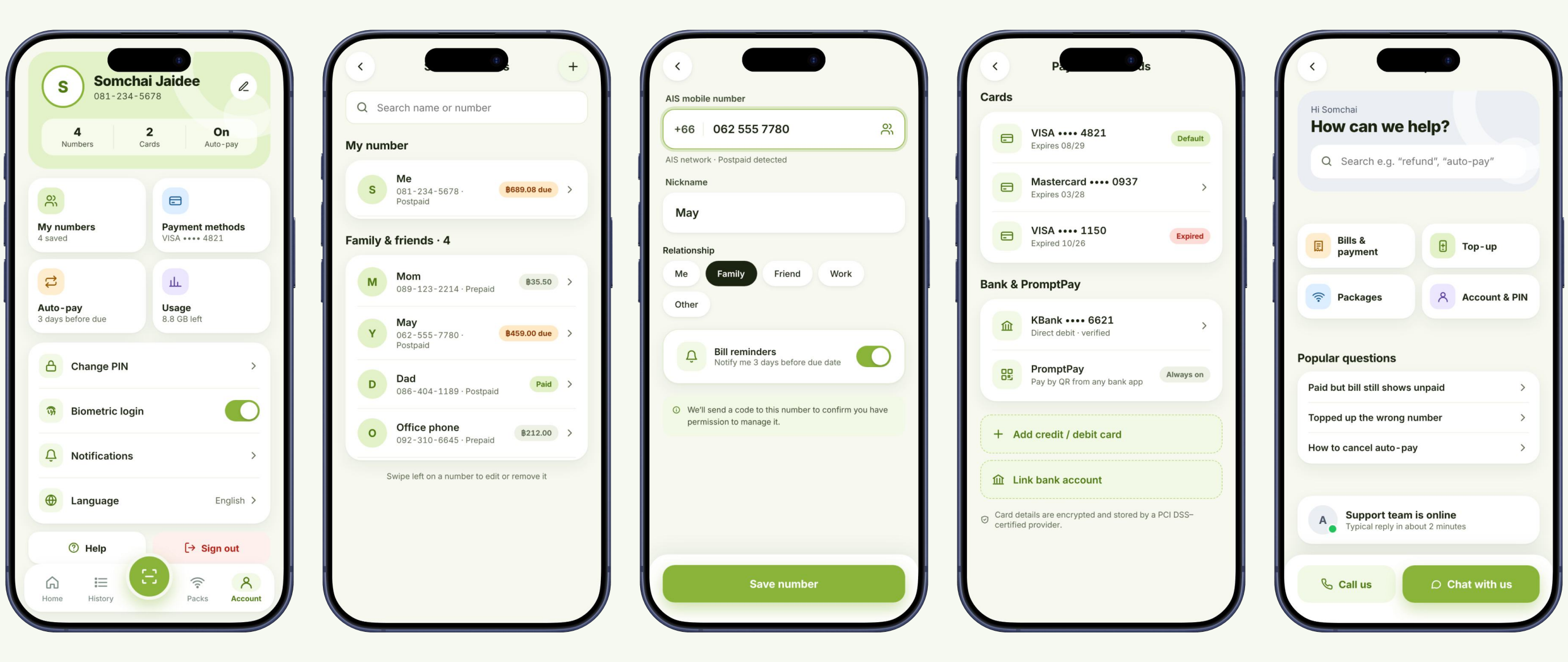

Create a 6-digit PIN
You'll use it to confirm every payment. Avoid birthdays or repeated numbers.
Step 1 of 2 · Enter PIN
Skip
Pay faster with biometrics
Use your fingerprint or face instead of typing your PIN. You can change this anytime in Account.
Your biometric data stays on this device
PIN is still available as a backup
Enable biometrics
Not now
Somchai
081-234-5678
Postpaid · 5G
This month's bill
฿689.08
Due Nov 25 · 17 days left
Pay bill
Top up
Pay bill
Data packs
History
Usage
Receipts
Auto-pay
Help
8.8 GB data left
of 60 GB · resets in 22 days
Add data
Quick top-up
See all
Mom
May
Dad
Office
Add
Home
History
Packs
Account
Top up to
089-123-2214
Mom · AIS Prepaid
Amount
฿100
Balance ฿35.50 → ฿135.50
Other amount (10 – 1,500)
Pay with
QR PromptPay
•• 4821 VISA
Bank Direct debit
Low-balance alert
Notify me when Mom has under ฿20
Total ฿100.00
Continue
Search name or number
All
AIS only
Favourites
Only AIS prepaid numbers can be topped up
Recent
Arthit P.
Anong S.
Bee Chanya
Other network
Dad
Kittipong
Kanya R.


Mobile number
095-118-2270
This number isn't on the AIS network
Select amount
Can't top up this number
Top-ups work only for AIS prepaid numbers. Check the number, or pick another contact.
Choose another contact →
Top up now
You're topping up
฿100.00
Mom · 089-123-2214
Network AIS Prepaid
Current balance ฿35.50
Balance after top-up ฿135.50
Service fee Free
Total ฿100.00
Pay with
PromptPay / QR
Change
Promo code
Apply
Save Mom as a favourite
Mom's balance updates within seconds
Confirm & pay ฿100.00
VISA
4821 •••• •••• 4821
CARD HOLDER SOMCHAI J.
EXPIRES 08/29
Card number
4821 1234 5678 4821
Expiry 08/29
CVV 3 digits
Name on card SOMCHAI J.
Save card for next time
We accept VISA MASTERCARD JCB
Pay ฿100.00
Secured with 3-D Secure · We never store your CVV
Enter your PIN
to confirm payment of ฿100.00 to 089-123-2214
Forgot PIN?
Top-up successful
Mom's new balance is ฿135.50
Mom 089-123-2214 · Prepaid ฿100
Paid with PromptPay
Date Nov 8, 2026 · 10:18 PM
Reference TU26110822181
Share
Save slip
Top up again
Done


History
081-234-5678 · Postpaid
Unpaid
Amount due
฿689.08
Due date Nov 25, 2026
Billing period Oct 1–31
Monthly plan 599.00
Extra 5 GB pack 45.00
VAT 7% 45.08
View full invoice
฿47.08 more than October
Mostly from the extra 5 GB pack on Oct 14
Pay with
Auto-pay is on
3 days before due date
Pay ฿689.08
Invoice no.
INV-2611-0812345
Account Somchai J.
Number 081-234-5678
Billing period Oct 1–31, 2026
Due date Nov 25, 2026
Monthly plan ฿599 599.00
Extra 5 GB pack · Oct 14 45.00
Calls over plan 0.00
Subtotal 644.00
VAT 7% 45.08
Total due ฿689.08
Request tax invoice
Full e-Tax invoice sent to your email
Question about a charge?
Chat with us about this invoice
PDF
PROMPTPAY
Amount
฿689.08
AIS Postpaid · 081-234-5678
Ref. 2611 0812 3456
Expires in 14:52
Save or screenshot the QR
Scan it in your bank app
Confirm — bill updates instantly
Save QR
I've paid
Payment successful
Your October bill is paid. Receipt sent by SMS.
Amount paid
฿689.08
Number 081-234-5678
Paid with PromptPay
Date Nov 8, 2026 · 10:16 PM
Transaction ID TX26110822164
Back to home
Share receipt
Payment didn't go through
Your bank declined the transaction. No money was taken from your account.
Amount ฿689.08
Method VISA •••• 4821
Reason Declined by issuer (05)
Time Nov 8, 2026 · 10:16 PM
What you can do
Check your card limit or balance
Try PromptPay or another card
Contact your bank if this keeps happening
Try again
Use another method
Contact support


November 2026 · 3 receipts
฿399.00
Download all
Tax invoice
Search receipts
All
Bills
Top-ups
Packs
NOVEMBER
10 GB · 30 days ฿199.00
Top up · Mom ฿100.00
Top up · Mom ฿100.00
OCTOBER
October bill ฿642.00
Top up · May ฿50.00
Top up · Mom ฿100.00
Auto-pay is on
081-234-5678 · Postpaid
Next payment Nov 22, 2026 ฿689.08
Pay from
VISA •••• 4821
Default card · expires 08/29
When to pay
On the due date
3 days before due date
When the bill is ready
Max amount per bill ฿ 1,000
Remind me before paying
Save settings
Packages
Data
Calls
Roaming
MOST POPULAR
10 GB
Full-speed 5G/4G · 30 days
฿199
Buy now
Unlimited 4 Mbps ฿150
Max speed day pass ฿39
5 GB ฿79
20 GB ฿299
10 GB
Full-speed 5G/4G data for 30 days
฿199 incl. VAT · one-time
Max speed
30 days
No renew
After you buy
18.8 GB available until Dec 8
Details
Used before your plan's data
Works with hotspot and tethering
Terms & conditions
For 081-234-5678 · Change
Buy for ฿199
Confirm purchase
Package 10 GB · 30 days
Number 081-234-5678
Starts Right away
Expires Dec 8, 2026 · 10:20 PM
How to pay
Add to monthly bill
Pay now
Total ฿199.00
Confirm


10 GB is ready to use
Your package is active on 081-234-5678
Package data 10 GB / 10 GB
Valid until Dec 8, 2026 · 10:20 PM
Charged to December bill
Amount ฿199.00
We'll remind you when 80% of the package is used and 1 day before it expires.
Done
View usage
8.8 GB left
Mobile data
51.2 / 60 GB used
85% used
Resets in 22 days
Calls 210 / 300 min
SMS 12 / 100
Data use · last 7 days
avg 2.6 GB/day
1.6 2.4 1.9 3.1 2.2 4.0 2.7
Mon Tue Wed Thu Fri Sat Sun
Plan
Monthly plan ฿599
Change
Extra 5 GB pack ฿45.00
Buy more data
History
Spent in November
฿399.00
3 payments · −฿243 vs Oct
November ฿399.00
Top up · Mom −฿100.00 Success
10 GB · 30 days −฿199.00 Success
Top up · Mom −฿100.00 Success
October ฿642.00
Package 10 GB ฿199.00 Failed
October bill −฿642.00 Success
Top up · Mom
฿100.00
Successful
To 089-123-2214
Network AIS Prepaid
Paid with PromptPay · KBank
Date Nov 8, 2026 · 10:18 PM
Reference TU26110822181
Status
Payment received
Sent to AIS
Balance updated
Report issue
Receipt
Mark all read
All
Bills
Top-ups
Offers
TODAY
Your November bill is ready
฿689.08 is due on Nov 25. Pay early to keep auto-pay on track.
2 family bills are due
Mom and May have unpaid postpaid bills. Pay them together.
Top-up successful
฿100.00 added to 089-xxx-2214 (Mom).
EARLIER
Data running low
You've used 85% of your 60 GB. Add a pack to keep full speed.
Payment failed
Your card was declined for ฿199.00. No money was taken.


Somchai Jaidee
081-234-5678
4 Numbers
2 Cards
On Auto-pay
My numbers
Payment methods
Auto-pay
Usage
Change PIN
Biometric login
Notifications
Language English
Help
Sign out
Search name or number
My number
Me 081-234-5678 · Postpaid ฿689.08 due
Family & friends · 4
Mom ฿35.50
May ฿459.00 due
Dad Paid
Office phone ฿212.00
Swipe left on a number to edit or remove it
AIS mobile number
+66 062 555 7780
AIS network · Postpaid detected
Nickname
May
Relationship
Me
Family
Friend
Work
Other
Bill reminders
Notify me 3 days before due date
We'll send a code to this number to confirm you have permission to manage it.
Save number
Cards
VISA •••• 4821 Default
Mastercard •••• 0937
VISA •••• 1150 Expired
Bank & PromptPay
KBank •••• 6621
PromptPay Always on
Add credit / debit card
Link bank account
Card details are encrypted and stored by a PCI DSS-certified provider.
Hi Somchai
How can we help?
Search e.g. "refund", "auto-pay"
Bills & payment
Top-up
Packages
Account & PIN
Popular questions
Paid but bill still shows unpaid
Topped up the wrong number
How to cancel auto-pay
Support team is online
Typical reply in about 2 minutes
Call us
Chat with us